UME

Organization / Business Operating System

Product Requirements Document

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Document Information & Revision History

Version	Date	Author	Status	Summary of Changes
0.1.0	Jan 2026	UME Product Team	Draft	Initial outline and persona research
0.5.0	Feb 2026	UME Product Team	Draft	Feature catalogue and user stories (first pass)
0.9.0	Mar 2026	UME Product Team	Review	NFRs, release plan, and acceptance criteria added
1.0.0	Mar 2026	UME Product Team	Approved	Baseline for Milestone 1 engineering kick-off

How to Use This Document

Product Managers — §3 User Stories, §6 Release Plan, §7 Success Metrics
Engineering Leads — §4 Feature Catalogue, §5 Non-Functional Requirements, §8 Dependencies
Design Team — §2 Personas, §3 User Stories, §5 Usability NFRs
QA / Test Engineers — §3 Acceptance Criteria, §5 NFRs, Appendix C (AC Template)
Executives / Board — Executive Summary, §1 Vision, §6 Roadmap, §7 Success Metrics

Executive Summary

UME is the world's first Organisation Operating System — a single, unified platform that replaces the fragmented stack of 10–30 disconnected tools that every organisation currently relies on. Built in Rust for performance and correctness, UME provides 42 domain modules covering every aspect of running an organisation: legal entity management, finance and accounting, human resources, governance, risk and compliance, sales, marketing, operations, supply chain, IT, security, and strategic management.

This Product Requirements Document (PRD) defines what UME must do as a product — its features, user stories, acceptance criteria, release prioritisation, and measurable outcomes. It is the authoritative reference for the product team, engineering, design, and QA.

42	18+	12	45+
Org Modules	**User Personas**	**Org Types Supported**	**Jurisdictions (Day 1)**
domain capabilities	*roles across all org types*	*legal structures*	*compliance policy packs*

Product Vision Statement

UME gives every organisation — from a solo entrepreneur to a global enterprise — a single operating system that holds their complete organisational context and provides every capability they need to govern, operate, and grow, without requiring them to integrate a stack of disconnected tools.

Theme	What UME Delivers
Unified Organisational Model	One platform that knows your structure, entities, obligations, people, money, and strategy — and keeps all of them consistent
Automated Compliance	Jurisdiction-aware legal entity management that derives filing obligations from organisational facts and alerts you before deadlines, not after
Financial Intelligence	Real-time consolidated financials across every entity; automated multi-entity reconciliation; proactive cash flow forecasting
Connected Risk	Risk indicators that update automatically from live operational data across HR, Finance, IT, and Operations
People-Centred Operations	HR lifecycle events that cascade automatically through Finance, Security, Process, and Compliance without manual coordination
Universal Accessibility	One platform that works correctly for a sole trader, a non-profit, a cooperative, an investment fund, or a government department
Extensible by Design	A module SDK that lets any organisation or third-party developer add domain-specific capabilities fully integrated with the platform

1. Product Vision & Strategy

1.1 Vision

Every organisation deserves an operating system. Not a collection of tools, not an ERP bolted together with APIs, not a suite of apps that share a logo but not a data model — an actual operating system that understands what the organisation is, what it is obligated to do, who works in it, and what it is trying to achieve.

UME is that operating system. Its kernel mediates all organisational data. Its 42 modules provide every functional capability. Its event bus ensures that a change in one domain — a new hire, a period lock, a filing due date, a risk threshold breach — automatically propagates to all affected domains without manual intervention.

1.2 Strategic Objectives

Objective	Measure of Success	Horizon
Eliminate organisational fragmentation	>80% of deployed orgs report UME as their only primary operational platform	12 months post-GA
Automate compliance globally	<5% of UME-deployed orgs experience a missed statutory deadline	18 months post-GA
Unify multi-entity financial management	Finance teams report consolidated group view available in <4 hours vs. days	12 months post-GA
Make GRC accessible at any scale	SMEs with no prior GRC practice successfully adopt risk management in UME	18 months post-GA
Build the organisational intelligence layer	Executives report UME provides their primary operational dashboard	24 months post-GA
Create an open ecosystem	50+ third-party custom modules available via the UME marketplace	24 months post-GA

1.3 Product Principles

1. Organisation-first: every feature is designed around the organisational model, not around a functional domain.
2. Event-driven correctness: state changes propagate automatically; no manual synchronisation between domains.
3. Compliance by default: legal and regulatory obligations are derived from organisational facts, not configured manually.
4. Scale without migration: the same platform works for 1 user and 100,000 users; no upgrade to a "bigger" product.
5. Accessible complexity: powerful capabilities (GRC, multi-entity consolidation) must be usable by non-specialists.
6. Audit as infrastructure: every operation is auditable from day one; compliance is not bolted on after.
7. Extensible core: the platform is a foundation; domain-specific depth is added through modules, not through rebuilding.

1.4 Positioning

Segment	Positioning
Solo Entrepreneurs & Freelancers	The first platform that manages your entity compliance, finances, contracts, and operations in one place — no accountant required for day-to-day management.
Small & Medium Businesses	Replace your stack of Xero + BambooHR + spreadsheets + your lawyer's reminder emails with one platform that does all of it correctly.
Growth & Scale-Up Companies	The operating system that grows with you: works for 10 people today and 500 tomorrow, without a migration or a painful ERP implementation.
Multi-Entity & Holding Structures	Consolidated group financials, cross-entity compliance, and unified risk management — without a team of analysts manually reconciling spreadsheets.
Non-Profits & NGOs	The only platform that understands restricted funds, grant lifecycle, trustee governance, and donor reporting as first-class capabilities.
Government & Public Sector	Appropriations compliance, procurement governance, FOI management, and public audit trail — built in, not bolted on.
Investment Funds	Portfolio company tracking, LP capital accounts, IC deal governance, and carry calculations alongside the fund's own operational HR and Finance.

2. User Personas & Jobs To Be Done

UME serves 18 primary user personas grouped into three tiers. Each persona is defined by their role, their primary job to be done in UME, and their core frustration with the status quo.

2.1 Persona Tier 1 — Executive & Board

Persona	Title	Primary JTBD	Status Quo Frustration
P-01 CEO / Founder	Chief Executive	See the complete state of the organisation — financial, people, risk, compliance — in one place, and make decisions with confidence.	I have 8 dashboards open and none of them tell me the same story about my business.
P-02 CFO / Finance Director	Chief Financial Officer	Close the books faster, produce consolidated group accounts, forecast cash, and answer board questions without a multi-day data collection exercise.	Month-end close takes 10 days because I'm manually pulling reports from 4 different accounting instances.
P-03 COO / Operations Director	Chief Operating Officer	Ensure every operational process runs reliably, escalations happen automatically, and I can see cross-departmental bottlenecks before they become crises.	I find out about process failures from an angry customer, not from my systems.
P-04 Board Director / Trustee	Non-Executive Director	Receive a complete, accurate board pack before each meeting, vote on resolutions with proper conflict-of-interest management, and maintain my statutory obligations.	Board packs arrive 24 hours before the meeting, in PDFs, with no way to ask questions about the underlying data.

2.2 Persona Tier 2 — Functional Leads

Persona	Title	Primary JTBD	Status Quo Frustration
P-05 CHRO / HR Director	Chief People Officer	Run the full employee lifecycle — hire to retire — with automatic downstream effects on Finance, IT, and Compliance.	Every new hire requires me to send emails to 6 departments to trigger their respective processes.
P-06 General Counsel / Legal Officer	Chief Legal Officer	Know every statutory obligation across every entity in the group, maintain the filing calendar, and have evidence ready for any audit or regulator.	I track compliance deadlines in a spreadsheet. The day I miss updating it is the day we miss a filing.
P-07 CRO / Risk Manager	Chief Risk Officer	Maintain a live risk register that updates automatically from operational KPIs, escalate to the board when appetite is breached.	Our risk register is last month's data. By the time we review it, the risk has already materialised.
P-08 CTO / IT Director	Chief Technology Officer	Manage the full IT asset and service lifecycle, respond to incidents with structured workflows, and demonstrate security compliance.	I have a CMDB in one tool, incidents in another, and changes in a third. Nothing talks to anything else.
P-09 CMO / Marketing Director	Chief Marketing Officer	Plan campaigns with strategic context, execute across channels, measure attribution precisely, and iterate based on live signals.	I know our campaign spent £80K. I don't know if it generated £80K of revenue

			— the data lives in 3 different systems.
P-10 VP Sales / Revenue Leader	VP of Sales	Manage pipeline from lead to close, forecast accurately, pay commissions correctly, and connect won revenue directly to Finance.	Our CRM says we closed £500K this quarter. Finance says £420K. Nobody can explain the difference.

2.3 Persona Tier 3 — Operational & Individual Contributors

Persona	Title	Primary JTBD	Status Quo Frustration
P-11 Finance Manager	Finance Manager	Process invoices, manage budgets, run payroll, reconcile accounts, and produce reports without manual data gathering.	I spend 3 days a month reconciling intercompany transactions that should be automatic.
P-12 HR Manager	HR Manager	Onboard and offboard employees smoothly, manage leave, track certifications, and run performance reviews without chasing people.	Offboarding takes a week because I have to email IT, Finance, and Legal separately to trigger their steps.
P-13 Compliance Manager	Compliance Manager	Track every regulatory obligation, collect evidence automatically, manage audit engagements, and never miss a deadline.	Evidence collection before an audit is 2 weeks of emailing colleagues for screenshots.
P-14 Project / Programme Manager	Project Manager	Track work from inception to completion with resource assignments, budgets, and milestone visibility across multiple projects.	My project tool doesn't know about budget actuals. Finance doesn't know about project milestones. Nobody has the full picture.
P-15 Knowledge Worker / Employee	Employee	Find information, request leave, submit expenses, complete training, and contribute to work items from a single portal.	I use 6 different apps to do my job. None of them remember my preferences. None of them talk to each other.
P-16 IT Administrator	IT Admin	Manage platform health, user access, module configuration, backups, and system changes with full audit trail.	Making a config change to our HRIS requires a support ticket. Verifying it was done right requires another one.
P-17 Independent Worker / Freelancer	Freelancer	Manage clients, proposals, contracts, time, invoices, and tax obligations from a single platform.	I use Notion for clients, Toggl for time, FreshBooks for invoices, and my accountant for tax. I'm paying for 4 tools for a 1-person business.
P-18 System Administrator	Sys Admin	Configure the UME platform, manage custom modules, control RBAC, and maintain infrastructure without specialist Rust knowledge.	Platform configuration requires editing TOML files and restarting services. There's no UI for anything operational.

3. User Stories & Acceptance Criteria

User stories are grouped by module domain. Each story is written in the standard format: "As a [persona], I want to [action] so that [outcome]." Acceptance criteria define the minimum conditions for the story to be considered complete.

3.1 Organisation Administration (MOD-01)

US-ADM-001 As P-01 CEO, I want to manage the organisation's entity structure so that I can see all entities in the group, their relationships, and their current compliance status in one view.	
Acceptance Criteria	1. Org unit tree is displayed with parent-child relationships and depth up to 10 levels 2. Each org unit shows: name, type, manager, headcount, and status 3. Restructuring (move org unit) is a single drag operation with audit trail 4. Deleting an org unit with active employees or children is prevented with a clear error message

US-ADM-002 As P-06 Legal Officer, I want to publish an organisation-wide policy and track acknowledgment so that every affected employee has formally acknowledged the policy and I have evidence for audit.	
Acceptance Criteria	1. Policy can be created with rich text, version number, and effective date 2. Target audience can be defined by org unit, role, or custom filter 3. Acknowledgment request is sent via the Comms module automatically on publish 4. Acknowledgment rate is visible in real time; overdue acknowledgments trigger automated reminders 5. Full acknowledgment evidence is exportable as a PDF for audit

3.2 Finance & Accounting (MOD-14)

US-FIN-001 As P-02 CFO, I want to see the consolidated financial position of the entire group in real time so that I can answer board-level questions about group cash, P&L, and balance sheet without waiting for month-end.	
Acceptance Criteria	1. Consolidated group P&L, balance sheet, and cash flow are available in real time, not just at period end 2. Intercompany eliminations are applied automatically; no manual adjustment required 3. Drill-down from group total to entity to individual journal entry is available in one click 4. Report exports to XLSX and PDF in GAAP/IFRS format

US-FIN-002 As P-11 Finance Manager, I want to process a supplier invoice through 3-way match so that invoices are only paid when they match an approved PO and goods receipt, with no manual intervention.	
Acceptance Criteria	1. Invoice can be created manually or ingested via email/API 2. System automatically matches invoice to PO and goods receipt by document reference or line-item matching 3. Match failure creates a workflow exception routed to the responsible buyer with details of the discrepancy 4. Matched invoices are queued for the next payment run automatically 5. Full match audit trail is visible on the invoice record

US-FIN-003 As P-11 Finance Manager, I want to run payroll for the current period so that all employees are paid correctly with accurate deductions and a journal entry is posted automatically.	
Acceptance Criteria	1. Payroll run calculates gross pay, statutory deductions, employer costs, and net pay for all active employees 2. Jurisdiction-specific tax and contribution rules are applied automatically based on employee location

	3. Payroll journal entry is generated and posted to the general ledger automatically on approval 4. Payslips are distributed to employees via the portal/comms module 5. Payroll run report is downloadable for HMRC/IRS/equivalent submission

US-FIN-004 As P-17 Freelancer, I want to send a professional invoice and track whether it has been paid so that I get paid on time and have a complete record of all client transactions for tax purposes.	
Acceptance Criteria	1. Invoice created from a template with client details, line items, tax, and payment terms in under 2 minutes 2. Invoice is emailed to the client directly from UME with a payment link 3. Payment status (sent, viewed, paid, overdue) is tracked automatically 4. Automated payment reminders sent at configurable intervals for overdue invoices 5. All invoices for the period are summarised for tax purposes with one click

3.3 Legal Entity Management — Chombo (MOD-13)

US-CHM-001 As P-06 Legal Officer, I want to see every upcoming statutory filing deadline across all entities so that no filing ever falls through the cracks regardless of jurisdiction or entity type.	
Acceptance Criteria	1. Filing calendar shows all obligations across all entities sorted by due date 2. Amber alert is triggered at the configured lead_days before each deadline (default 30 days) 3. Red alert and escalation email is sent when a deadline is within 7 days with no evidence filed 4. Overdue filings turn red immediately and trigger escalation to the entity's designated executive 5. Evidence attachment and "mark as filed" workflow is available for each filing item

US-CHM-002 As P-01 Founder, I want to register a new legal entity and immediately know what I need to do so that I can set up a new subsidiary and have a complete compliance picture in under 10 minutes.	
Acceptance Criteria	1. New entity is created by entering name, type, jurisdiction, and registration number 2. System immediately evaluates applicable policy packs and generates the filing schedule 3. First-year compliance obligations are displayed in priority order 4. Entity appears in the group structure automatically 5. Notification is sent to the designated compliance manager

3.4 Human Resources (MOD-16)

US-HR-001 As P-12 HR Manager, I want to onboard a new employee so that the new hire has their access, equipment, training, and welcome pack on day one without me chasing anyone.	
Acceptance Criteria	1. Creating an employee record automatically triggers the onboarding workflow in the Process module 2. Onboarding workflow includes: IT access provisioning task, equipment request task, policy acknowledgment, and mandatory training assignments 3. Each task is assigned to the responsible person automatically based on org unit and role 4. Manager receives a checklist of their onboarding responsibilities via the portal 5. New hire has access to the employee portal and their onboarding plan on start date 6. Onboarding completion status is visible to HR manager in real time

US-HR-002 As P-15 Employee, I want to request annual leave and see my balance so that I know exactly how much leave I have and my request is approved (or declined) without paperwork.	
Acceptance Criteria	1. Employee sees current leave balance for all leave types on their portal home screen 2. Leave request is submitted in under 30 seconds: select type, dates, optional note 3. Request routes automatically to the configured approver (usually line manager) 4. Approved leave blocks the employee's calendar in the Schedule module

	5. Leave balances update in real time on approval 6. Email notification sent to employee on approval or decline with reason

US-HR-003 As P-05 CHRO, I want to run the annual performance review cycle so that every employee has a completed review by the end of the review period without me manually tracking progress.	
Acceptance Criteria	1. Review cycle is created with: review period, participant scope, self-assessment deadline, manager review deadline, calibration session date 2. Self-assessment form is sent to all in-scope employees automatically on cycle open date 3. Manager review is triggered automatically when employee self-assessment is submitted 4. Completion dashboard shows real-time progress: % self-assessments submitted, % manager reviews complete 5. Overdue reviews trigger automated reminders to the employee/manager 6. Final ratings are locked after the calibration session; employees receive their outcome notification

3.5 Governance, Risk & Compliance (MOD-15 & MOD-33)

US-GRC-001 As P-07 Risk Manager, I want to know immediately when a key risk indicator breaches its threshold so that risks are escalated to the right people automatically; no weekly manual check required.	
Acceptance Criteria	1. KRIs are configurable with amber and red thresholds per risk 2. KRI values update automatically from connected data sources (Finance KPIs, HR metrics, IT incidents) 3. Amber breach triggers an email/portal notification to the risk owner within 5 minutes of data update 4. Red breach triggers escalation to the executive sponsor and Risk Committee chair automatically 5. All KRI alerts are recorded in the audit log with timestamp and value at alert time

US-GRC-002 As P-13 Compliance Manager, I want to prepare for an internal audit without 2 weeks of evidence collection so that audit evidence is always current and automatically linked to controls; preparation takes hours, not weeks.	
Acceptance Criteria	1. Each control in the control library has an assigned evidence collection method (manual, automated, or both) 2. Automated evidence is pulled from linked modules (e.g., access review evidence from Security, training completion from Learning) 3. Audit engagement is created with scope, timeline, and assigned reviewers 4. Evidence package for each control is generated automatically from linked evidence sources 5. Findings are recorded against controls with management responses and remediation deadlines 6. Audit report is generated from findings with one click

3.6 Sales & CRM (MOD-34 & MOD-08)

US-SAL-001 As P-10 VP Sales, I want to forecast revenue accurately for the next quarter so that I can commit to a number with confidence, not a guess based on hope and gut feel.	
Acceptance Criteria	1. Pipeline view shows all opportunities by stage with probability-weighted value 2. Weighted forecast is calculated automatically from stage probability × deal value 3. AI-assisted deal score is visible per opportunity based on historical win/loss patterns 4. Forecast vs. target gap is visible at the top of the pipeline view 5. Forecast can be sliced by rep, territory, vertical, and time period 6. Forecast history is tracked so forecast accuracy can be measured over time

US-SAL-002 As P-10 VP Sales, I want to calculate and communicate commission statements automatically so that reps trust their commission figures; no disputes; no manual spreadsheets.	
Acceptance Criteria	1. Commission plan is configured once per plan period with rules, rates, accelerators, and quotas 2. Commission is calculated automatically on opportunity close linked to Finance 3. Statement is generated per rep per period showing: quota, achievement, rate, amount earned 4. Statements are distributed to reps via the portal with the ability to dispute individual items 5. Approved commissions are passed to the Finance module payroll integration automatically

3.7 Security, Privacy & IT (MOD-36 & MOD-18)

US-SEC-001 As P-08 CTO, I want to respond to a P1 security incident with a structured, audited process so that the incident is contained and resolved within SLA; a complete post-incident report is available.	
Acceptance Criteria	1. Incident is created with classification (P1–P4), affected systems, and initial description 2. P1 declaration triggers immediate notifications to CISO, CTO, and on-call team 3. SLA timer starts on incident creation; countdown is visible on the incident record 4. Incident timeline records every action, by whom, with timestamps 5. Containment, eradication, and recovery steps are tracked with evidence 6. Post-Incident Review (PIR) template is auto-populated from the incident timeline 7. PIR action items create Work items assigned to responsible owners

US-IT-001 As P-16 IT Admin, I want to manage a change to a production system through a full CAB process so that no unauthorised changes reach production; every change is documented and reversible.	
Acceptance Criteria	1. Change request captures: description, affected CIs, risk assessment, rollback plan, and proposed schedule 2. Standard changes auto-approve; Normal changes route to CAB members for review 3. Emergency changes route to emergency approver with 2-hour SLA 4. Approved changes appear in the change calendar for the implementation window 5. Post-implementation review records outcome (successful, failed, rolled back) with evidence 6. All change records are linked to the affected CMDB configuration items

3.8 Work Management & Portal (MOD-40 & MOD-26)

US-WRK-001 As P-14 Project Manager, I want to run a sprint with full visibility into team capacity and delivery progress so that I can tell stakeholders exactly where we are, what is at risk, and when we will deliver.	
Acceptance Criteria	1. Sprint is created with goal, start/end dates, and work items pulled from the backlog 2. Work items are assigned with story points and time estimates 3. Daily burn-down and velocity charts update automatically as items are completed 4. Items blocked or at risk are flagged; automatic notification to PM when item SLA is at risk 5. Sprint review summary is auto-generated from completed items with linked evidence

US-WRK-002 As P-15 Employee, I want to find any organisational information and take action from a single portal so that I never need to open another tool for day-to-day work.	
Acceptance Criteria	1. Global search returns results from all modules: people, documents, work items, knowledge articles, policies 2. Notification centre consolidates alerts from all modules in one feed with action buttons 3. Quick actions allow: request leave, log time, submit expense, create work item — all in under 30 seconds 4. My work dashboard shows: tasks due today, approvals pending, upcoming leave, learning completions due 5. Module navigation is role-aware: only modules relevant to the user's role are shown

4. Feature Catalogue

Priority Key

P0 — Must-have for launch. Absence prevents the product from being usable for its core purpose.
P1 — Required for general availability. Absence significantly limits value for key personas.
P2 — Important for growth. Absence constrains specific segments but does not block core use.
P3 — Desirable. Adds value and competitive differentiation; acceptable to defer post-launch.

4.1 Platform Kernel & Infrastructure

Feature ID	Feature	Description	Pri
F-KRN-001	Kernel Lifecycle	Single-process kernel that boots all active modules in dependency order within 30 seconds; graceful shutdown with draining	P0
F-KRN-002	Module Registry	Dynamic module registration and discovery; dependency graph validation; hot-reload without full restart	P0
F-KRN-003	RBAC Engine	Role-based access control with role inheritance, resource scoping, and per-permission auditing; <10ms decision time	P0
F-KRN-004	Event Bus	Publish-subscribe event bus with glob pattern subscriptions, backpressure, dead-letter queue, and idempotent delivery	P0
F-KRN-005	Audit Chain	Immutable, SHA-256 cryptographically chained audit log; every write operation produces an audit record; 7-year retention	P0
F-KRN-006	Storage Abstraction	Unified storage API across SQLite (embedded), PostgreSQL (standard/enterprise), with Redis hot-cache tier	P0
F-KRN-007	Supervisor Engine	Per-module health monitoring at configurable interval; exponential backoff restart policy; fault isolation	P0
F-KRN-008	Executor Pool	8 named thread pools with per-pool resource limits; prevents any domain from starving others	P0
F-KRN-009	Schema Migration	Versioned, idempotent, forward-only database migrations; auto-applied on boot; checksum verification	P0
F-KRN-010	Device Driver Bus	Hot-swappable drivers for storage, identity, comms, broker, crypto, and search; circuit-breaker per driver	P0
F-KRN-011	Memory Manager	Named memory regions with configurable eviction policies; prevents unbounded memory growth	P1
F-KRN-012	Job Scheduler	Cron-based and one-shot scheduled jobs; bound to executor pool; job history and DLQ	P1

4.2 Security & Identity (MOD-36)

Feature ID	Feature	Description	Pri
F-SEC-001	Local Authentication	Username/password with bcrypt (cost 12) or Argon2id; account lockout after 5 failed attempts; 15-minute lockout window	P0

F-SEC-002	SSO Integration	OAuth2/OIDC and SAML 2.0 integration for enterprise identity providers (Azure AD, Okta, Google Workspace)	P1
F-SEC-003	Multi-Factor Authentication	TOTP (Google Authenticator / Authy compatible) and FIDO2/WebAuthn (hardware key) support; MFA mandatory for admin roles	P0
F-SEC-004	JWT Token Management	RS256-signed JWTs; 60-minute access tokens; 7-day refresh tokens; token revocation via session store	P0
F-SEC-005	Cryptographic Vault	AES-256-GCM at-rest encryption; key rotation without downtime; HSM-compatible vault interface	P0
F-SEC-006	Incident Management	P1–P4 incident lifecycle with SLA timers, timeline recording, containment steps, and auto-generated PIR	P1
F-SEC-007	Vulnerability Management	Vulnerability register with CVSS scoring, asset linkage, remediation tracking, and SLA monitoring	P1
F-SEC-008	Privacy Module	Record of Processing Activities (RoPA), DPIA workflow, DSAR management, and consent registry	P1
F-SEC-009	SIEM Integration	Real-time security event forwarding via webhook to SIEM platforms (Splunk, Elastic, Datadog)	P2
F-SEC-010	IP Allowlisting	Per-organisation IP allowlist enforcement at the API layer	P2

4.3 Administration (MOD-01)

Feature ID	Feature	Description	Pri
F-ADM-001	Org Unit Hierarchy	Unlimited-depth organisational structure tree; drag-and-drop restructuring; full history of structural changes	P0
F-ADM-002	Policy Management	Rich-text policy authoring, versioning, approval workflow, publish/retire lifecycle, and acknowledgment tracking	P0
F-ADM-003	Enterprise Calendar	Multi-jurisdiction public holiday management; integration with Schedule and HR leave modules	P1
F-ADM-004	Delegation Management	Formal authority delegation with start/end dates, scope definition, and audit trail	P1
F-ADM-005	Notification Templates	Organisation-branded email and in-app notification templates; per-event customisation	P2

4.4 Legal Entity Management — Chombo (MOD-13)

Feature ID	Feature	Description	Pri
F-CHM-001	Entity Registry	Register legal entities of any type (corporation, LLP, non-profit, trust, etc.) with full statutory details	P0
F-CHM-002	Group Structure	Multi-entity group visualisation with ownership percentages and group entity relationship graph	P0
F-CHM-003	Jurisdiction Policy Packs	45+ jurisdiction compliance rule packs; obligations derived automatically from entity type and jurisdiction	P0
F-CHM-004	Filing Calendar	Automated filing schedule generation; configurable lead-time alerts; overdue escalation with evidence attachment	P0

F-CHM-005	UBO/PSC Register	Ultimate Beneficial Owner and Persons with Significant Control register with event-triggered update prompts	P1
F-CHM-006	Compliance Evaluation	On-demand and scheduled evaluation of entity compliance posture; alert summary dashboard	P0
F-CHM-007	Director/Officer Registry	Track appointments, resignations, and statutory obligations for all directors and officers across all entities	P1

4.5 Finance & Accounting (MOD-14)

Feature ID	Feature	Description	Pri
F-FIN-001	Double-Entry Ledger	Full double-entry accounting with chart of accounts, journal entries, and period management	P0
F-FIN-002	Multi-Entity Support	Separate ledgers per entity with shared chart of accounts template and automatic intercompany elimination	P0
F-FIN-003	Multi-Currency	Functional and presentation currency support; real-time FX rates; revaluation journals	P0
F-FIN-004	Financial Statements	P&L, balance sheet, cash flow statement in GAAP/IFRS format; consolidated group statements	P0
F-FIN-005	Accounts Payable	Supplier invoice creation, approval workflow, 3-way PO matching, and payment run management	P0
F-FIN-006	Accounts Receivable	Customer invoicing, payment terms, automated reminders, receipt allocation, and AR ageing report	P0
F-FIN-007	Payroll Integration	Payroll computation with jurisdiction-aware deductions; automatic journal posting; payslip generation	P0
F-FIN-008	Budget Management	Budget creation per cost centre/project; budget vs. actual variance reporting; variance alerts	P1
F-FIN-009	Restricted Fund Accounting	Fund-level P&L and balance sheet for non-profits; restriction management; fund utilisation reporting	P1
F-FIN-010	Group Consolidation	Automated group consolidation with intercompany elimination; minority interest calculation	P1
F-FIN-011	Cash Flow Forecasting	13-week rolling cash flow forecast incorporating AR, AP, payroll, and committed contracts	P1
F-FIN-012	Banking Integration	Open banking API integration for bank feed reconciliation; bank statement import	P2
F-FIN-013	Project Accounting	Cost allocation to projects; project P&L; capitalisation workflow	P2

4.6 Human Resources (MOD-16)

Feature ID	Feature	Description	Pri
F-HR-001	Employee Lifecycle	Complete lifecycle management: pre-hire through termination with status-driven workflow triggers	P0
F-HR-002	Compensation Management	Salary, hourly, and commission compensation records with effective dating and history	P0
F-HR-003	Leave Management	Configurable leave types with accrual policies; approval workflow; calendar integration; balance tracking	P0

F-HR-004	Recruitment	Job requisition, application tracking, interview scheduling, offer management, and onboarding handoff	P1
F-HR-005	Payroll Computation	Gross-to-net payroll calculation with jurisdiction-aware statutory deductions and employer costs	P0
F-HR-006	Performance Reviews	Configurable review cycles: self-assessment, manager review, calibration, and outcome communication	P1
F-HR-007	Certifications & Training	Mandatory certification tracking with expiry alerts; training completion linked to Learning module	P1
F-HR-008	Succession Planning	Key person identification, successor mapping, and readiness scoring	P2
F-HR-009	Workforce Analytics	Headcount trends, attrition, gender pay gap, and organisational structure health metrics	P2

4.7 Governance, Risk & Compliance (MOD-15)

Feature ID	Feature	Description	Pri
F-GRC-001	Obligation Register	Regulatory and contractual obligation inventory with jurisdiction tagging and ownership assignment	P0
F-GRC-002	Control Library	Control register linked to obligations; control testing with evidence; effectiveness rating	P0
F-GRC-003	Audit Management	Internal and external audit engagement lifecycle; finding management; management response tracking	P1
F-GRC-004	Automated Evidence	Automatic evidence collection from connected modules (access logs, training records, financial approvals)	P1
F-GRC-005	Regulatory Change Management	Regulatory change tracking; impact assessment workflow; control update queue	P2

4.8 Risk Management (MOD-33)

Feature ID	Feature	Description	Pri
F-RSK-001	Risk Register	Risk registry with category, likelihood/impact scoring, treatment type, and ownership	P0
F-RSK-002	KRI Monitoring	Key Risk Indicator library with automated value updates from Finance, HR, and IT data sources	P0
F-RSK-003	Threshold Alerting	Amber/red KRI threshold alerts with automatic escalation to risk owner and executive sponsor	P0
F-RSK-004	Treatment Plans	Treatment plan management with milestones, owners, and residual risk tracking	P1
F-RSK-005	Risk Heat Map	Visual 5×5 likelihood/impact heat map with drill-down to individual risks	P1
F-RSK-006	Risk Appetite Framework	Configurable risk appetite statements per category; automatic appetite breach detection	P1

4.9 Analytics & Reporting (MOD-02)

Feature ID	Feature	Description	Pri

F-ANL-001	Executive Dashboard	Real-time cross-domain KPI dashboard; configurable metric selection; <60s data refresh	P0
F-ANL-002	Cross-Domain KPIs	KPIs that combine data from multiple modules: revenue per employee, compliance score, cash runway	P0
F-ANL-003	Trend Analysis	3-year historical trend charts for all key metrics; anomaly detection with alert	P1
F-ANL-004	Custom Reports	Report builder with configurable dimensions, filters, and visualisations; scheduled delivery	P1
F-ANL-005	Export	Export all reports to CSV, XLSX, and PDF; automated delivery via email or webhook	P1
F-ANL-006	BI Integration	Data connector for Tableau, Power BI, Looker, and Metabase via JDBC/ODBC or REST export	P2

4.10 Sales & CRM (MOD-34 & MOD-08)

Feature ID	Feature	Description	Pri
F-SAL-001	Opportunity Pipeline	Lead-to-close opportunity management with configurable stages and probability weighting	P0
F-SAL-002	Revenue Forecasting	Weighted and commit-based forecast views; rep, team, and company level; forecast vs. target	P1
F-SAL-003	Commission Engine	Configurable commission plans with quotas, rates, accelerators, and SPIFF; automated calculation	P1
F-SAL-004	Territory Management	Territory definition and assignment; rep-level pipeline filtering	P2
F-SAL-005	AI Deal Scoring	ML-based deal health score from historical win/loss data; risk flag on stalled deals	P2
F-CRM-001	Account & Contact Management	Full account hierarchy with contacts, interactions, and relationship strength tracking	P0
F-CRM-002	Interaction Logging	Log calls, emails, and meetings; link to opportunities; automated interaction capture via email integration	P1

4.11 Operations & Supply Chain (MOD-25 & MOD-37)

Feature ID	Feature	Description	Pri
F-OPS-001	Procurement Lifecycle	Purchase requisition → PO → goods receipt → invoice → payment full lifecycle	P0
F-OPS-002	Inventory Management	Inventory ledger with configurable reorder triggers; multi-location support; stock take workflow	P1
F-OPS-003	Supplier Registry	Supplier master with risk rating, performance scorecard, and contact management	P1
F-OPS-004	Process Orchestration	Configurable workflow engine: human tasks, automated tasks, conditional branching, and event triggers	P1
F-OPS-005	Work Management	Full work item lifecycle: backlog, sprint/kanban, time logging, dependencies, and reporting	P0

4.12 Marketing — Soko (MOD-22)

Feature ID	Feature	Description	Pri
F-MKT-001	Campaign Management	Campaign lifecycle from brief to execution; multi-channel; budget tracking; KPI targets	P1
F-MKT-002	Strategy Pack Engine	70+ marketing strategy packs evaluated against campaign context; ranked insight and action recommendations	P1
F-MKT-003	Market Signal Ingestion	Real-time ingestion of market signals (clicks, conversions, impressions) from connected channels	P1
F-MKT-004	Attribution Reporting	CAC, CVR, CTR, ROI, and ROAS per campaign, channel, and audience segment	P1
F-MKT-005	CRM Integration	Closed-loop attribution: link won opportunities to originating campaigns	P2

4.13 IT & Asset Management (MOD-18)

Feature ID	Feature	Description	Pri
F-IT-001	CMDB	Configuration item registry for hardware, software, cloud resources, and services; automated discovery integration	P1
F-IT-002	ITSM	Incident, problem, and service request lifecycle with SLA management and escalation	P1
F-IT-003	Change Management	Change request with CAB review, emergency approval path, and post-implementation review	P1
F-IT-004	Software Licensing	License inventory, compliance monitoring, and renewal calendar	P2

4.14 Knowledge, Learning & Portal (MOD-19, MOD-20, MOD-26)

Feature ID	Feature	Description	Pri
F-KNL-001	Knowledge Base	Searchable knowledge articles with full-text search, tagging, and AI-assisted search	P1
F-LRN-001	Learning Paths	Structured learning paths with courses, quizzes, and completion certificates	P1
F-LRN-002	Compliance Training	Mandatory training assignment with deadline tracking and completion evidence for GRC module	P1
F-PRT-001	Employee Portal	Personalised portal home with tasks, notifications, quick actions, and module navigation	P0
F-PRT-002	Global Search	Cross-module full-text search returning people, documents, work items, and knowledge articles	P1
F-PRT-003	Notification Centre	Consolidated notification feed with action buttons; configurable notification preferences	P0

4.15 API, SDK & Clients

Feature ID	Feature	Description	Pri

F-API-001	REST API	Full RESTful API with OpenAPI 3.1 specification; cursor pagination; structured error responses	P0
F-API-002	WebSocket Events	Real-time event stream via WebSocket for dashboard updates and notifications	P1
F-API-003	Rate Limiting	Per-client API rate limiting with configurable limits; 429 response with Retry-After header	P0
F-API-004	Webhook Outbound	Configurable outbound webhooks on any event topic with HMAC-SHA256 signing and retry	P1
F-API-005	Java GUI Client	Full JavaFX GUI with screens for all 42 modules; offline read-only mode; deep-linking	P1
F-API-006	Java Shell Client	Menu-driven CLI client for shell and scripted access to all modules	P1
F-API-007	Java Watch Client	Real-time terminal dashboard for live KPI monitoring via WebSocket	P2
F-API-008	Module SDK	Rust SDK with MockKernel, scaffold generator, schema validator, and integration test harness	P0
F-API-009	GraphQL (Experimental)	GraphQL read API for complex client queries; experimental; opt-in	P3

4.16 Deployment & Operations

Feature ID	Feature	Description	Pri
F-DEP-001	Embedded Mode	Single binary with SQLite; zero external dependencies; 1-command local start	P0
F-DEP-002	Docker Compose Mode	Docker Compose stack with PostgreSQL and Redis; production-ready for up to ~500 users	P0
F-DEP-003	Kubernetes / Helm	Helm chart with HPA, PDB, and rolling upgrade support; targets enterprise scale	P1
F-DEP-004	Environment Config	All configuration via environment variables; no restart required for non-kernel config changes	P0
F-DEP-005	Backup & Recovery	Scheduled full, incremental, and differential backups; AES-256 encryption; point-in-time restore	P0
F-DEP-006	Health Endpoint	/health endpoint with per-module status; Prometheus /metrics endpoint	P0
F-DEP-007	Rolling Upgrade	Zero-downtime rolling upgrade; mixed schema version tolerance during rollout	P1
F-DEP-008	SBOM & Provenance	CycloneDX/SPDX software bill of materials; GPG-signed release artefacts	P2

5. Non-Functional Requirements

Non-functional requirements define the quality attributes of the UME platform. All P0 NFRs are mandatory for general availability. P1 NFRs must be met within 3 months of GA.

5.1 Performance

Metric	Target	Measurement Method	Priority
API response time (P50)	< 100ms	Prometheus histogram; measured at API gateway	P0
API response time (P95)	< 500ms	Prometheus histogram; measured at API gateway	P0
API response time (P99)	< 2,000ms	Prometheus histogram; measured at API gateway	P0
Executive dashboard refresh	< 60 seconds	Synthetic monitor; full KPI reload time	P0
Full-text search response	< 1 second	Measured in integration tests with 100K document corpus	P1
Batch report generation	< 30 seconds for standard reports	Automated report generation benchmark	P1
Event bus throughput	> 10,000 events/second (in-process)	Load test with simulated event flood	P1
Audit record write latency	< 50ms P99	Prometheus histogram on audit write path	P0
Kernel boot time	< 30 seconds with all 42 modules	CI boot benchmark	P0
Module health check	< 500ms per module	Supervisor engine timeout measurement	P0

5.2 Scalability

Dimension	Target	Notes
Concurrent users per node	500	Sustained concurrent active sessions per server node
Total employees per organisation	100,000	Employee records across all entities in the group
Work items per organisation	1,000,000	Total active and historical work items
Journal entries per organisation	10,000,000	Cumulative general ledger entries
CRM contacts per organisation	50,000	Active contact records
Horizontal scaling	Stateless; scale by adding nodes	Session state in Redis; shared PostgreSQL
Database read scaling	Read replicas for analytics queries	Write to primary; analytics queries to replica
Kubernetes auto-scaling	HPA triggers at 70% CPU	Custom metric: event bus queue depth

5.3 Availability & Reliability

Metric	Standard Tier	Enterprise Tier	Measurement
Uptime SLA	99.9% (8.7 hrs downtime/yr)	99.95% (4.4 hrs downtime/yr)	Synthetic monitor; 1-minute polling
RTO (Recovery Time Objective)	< 4 hours	< 1 hour	Measured during DR drills
RPO (Recovery Point Objective)	< 1 hour	< 15 minutes	Backup frequency verification
Planned maintenance window	< 2 hours/month	< 30 minutes/month	Measured against maintenance schedule
MTTD (Mean Time to Detect)	< 5 minutes	< 2 minutes	Alert pipeline latency measurement
MTTR (Mean Time to Resolve)	< 30 minutes	< 15 minutes	Incident management records
Data durability	≥ 99.999999% (8 nines)	≥ 99.999999% (8 nines)	S3/equivalent SLA for backup storage

5.4 Security

Requirement	Specification	Priority
Encryption at rest	AES-256-GCM for all persisted data; key managed via CryptoVault; rotation without downtime	P0
Encryption in transit	TLS 1.2+ mandatory for all API connections in production; TLS 1.3 preferred	P0
Authentication strength	bcrypt cost 12 or Argon2id for passwords; RS256 JWT; MFA mandatory for admin roles	P0
RBAC enforcement	All operations checked by kernel RBAC before execution; no bypass path exists	P0
Audit completeness	Every mutating operation produces an audit record; RBAC denials always audited	P0
PII handling	PII fields encrypted at field level; masked in logs; GDPR/CCPA-compliant data subject workflows	P0
Vulnerability management	SAST in CI; dependency scanning; container scanning; 72-hour SLA for critical CVEs	P0
Penetration testing	Annual third-party penetration test; critical findings remediated within 30 days	P1
SBOM	Software bill of materials published with every release (CycloneDX format)	P1
Session security	JWT revocation within 60 seconds; IP allowlisting option; concurrent session limit	P1

5.5 Usability & Accessibility

Requirement	Specification	Priority
System Usability Score	SUS score ≥ 75 ("Good") across all primary personas	P0
Task completion time	Time for a new user to complete core tasks (create employee, post journal, file evidence) < 5 minutes each	P1

Error messages	All error messages in plain language; include the cause and the corrective action; no technical jargon	P0
Accessibility standard	WCAG 2.1 Level AA compliance for the web portal and Java GUI client	P1
Responsive design	Web portal functional on screens ≥ 768px wide	P1
Internationalisation	UI supports right-to-left languages; date, time, and number formats locale-aware	P2
Onboarding	A new organisation can complete setup and post its first journal entry in < 30 minutes without documentation	P1

5.6 Maintainability & Quality

Requirement	Specification	Priority
Test coverage — ume_core, kernel	≥ 80% line coverage maintained in CI	P0
Test coverage — modules	≥ 70% line coverage maintained in CI	P0
Integration tests	Full module integration test suite using MockKernel; run on every PR	P0
Linting & formatting	cargo clippy and cargo fmt enforced in CI; zero warnings policy	P0
Dependency hygiene	Dependabot or equivalent; critical CVE patches within 72 hours	P0
Build time	Full workspace Rust build < 20 minutes in CI; Java build < 10 minutes	P1
Deployment automation	Zero-downtime rolling deploys; schema migration auto-applied on startup	P0
Deprecation window	Stable APIs deprecated for a minimum of 2 minor versions before removal	P0

6. Release Plan & Roadmap

UME is released in four milestones. Each milestone delivers a complete, usable increment of value — not a partial or preview release. Priority tags from the Feature Catalogue (§4) map directly to milestones.

Release Philosophy

Each release is a working product, not a checkpoint. Every milestone delivers end-to-end value for a defined set of personas.
P0 features ship in Milestone 1. P1 features ship in Milestones 1–2. P2 features ship in Milestones 3–4. P3 features are post-roadmap.

Milestone 1 — Foundation — Q3 2026 — Target: General Availability	
Theme	Complete end-to-end operational capability for a single-entity organisation.
Kernel	Full kernel lifecycle: RBAC, event bus, audit chain, storage abstraction, supervisor, executor pool.
Security	Local auth + MFA (TOTP); JWT tokens; cryptographic vault; AES-256 encryption.
Administration	Org unit hierarchy; policy management with acknowledgment tracking.
Finance	Full double-entry ledger; AP/AR; multi-currency; payroll; financial statements; period management.
HR	Employee lifecycle; leave management; payroll integration; compensation records.
Legal Entity	Entity registry; 20 priority jurisdiction policy packs; filing calendar with alerts.
Risk	Risk register; KRI monitoring with threshold alerts; basic heat map.
Work	Work items; sprint management; time logging; employee portal with notifications.
API	Full REST API; OpenAPI 3.1 spec; WebSocket events; rate limiting.
Deployment	Embedded mode (SQLite); Docker Compose mode; /health and /metrics endpoints.
SDK	Module SDK with MockKernel; scaffold generator; schema validator.
Quality Gate	SUS ≥ 75; P50 API < 100ms; P95 < 500ms; 80%+ kernel/core test coverage; zero known P0 bugs.
Milestone 2 — Growth — Q4 2026	
Theme	Multi-entity group operations; GRC capability; sales and marketing foundation.
Finance	Multi-entity consolidation; intercompany elimination; budget management; restricted funds; cash flow forecasting.
Legal Entity	Remaining 25+ jurisdiction policy packs; UBO/PSC register; director/officer registry.
GRC	Obligation register; control library; audit management; automated evidence collection.
Sales & CRM	Opportunity pipeline; revenue forecasting; account and contact management; commission engine.
Marketing (Soko)	Campaign management; strategy pack engine (70+ packs); market signal ingestion; attribution.
HR	Recruitment; performance reviews; certification and training tracking.
Security	Incident management; vulnerability register; privacy module (RoPA/DPIA/DSAR).
IT	CMDB; ITSM; change management.
Analytics	Cross-domain KPI dashboard; trend analysis; custom report builder; export.

Java Client	Java GUI client (all 42 module screens); Java shell client.
Deployment	Kubernetes Helm chart; rolling upgrade support.
Integrations	SSO (OAuth2/OIDC, SAML); outbound webhooks; banking integration (open banking).
Quality Gate	All P1 NFRs met; SUS ≥ 78; 500 concurrent user load test passing.
Milestone 3 — Ecosystem — Q1 2027	
Theme	Third-party extensibility; advanced analytics; operational depth.
SDK	Public custom module marketplace; module signing and verification; vendor registration.
Analytics	BI tool integration (Tableau, Power BI, Looker); anomaly detection.
Operations	Full supply chain module; project accounting; multi-echelon inventory.
Board	Board meeting management; resolution recording; board pack assembly.
ESG	ESG framework management; scope 1/2/3 emissions; social metrics; governance reporting.
Investment	Fund management; LP capital accounts; portfolio company tracking; IC governance.
Learning	Full LMS with learning paths, quizzes, and compliance training certificates.
Security	SIEM integration; IP allowlisting; automated security event forwarding.
Deployment	Terraform/Pulumi IaC modules; SBOM publication; GPG-signed releases.
Java Client	Java Watch client (live dashboard); deep-linking (ume://); offline read-only mode.
Milestone 4 — Intelligence — Q2 2027	
Theme	AI-assisted organisational intelligence; full sector coverage; enterprise depth.
AI Features	AI-assisted deal scoring; anomaly detection alerts; AI knowledge search; payroll anomaly detection.
GraphQL	GraphQL read API (experimental); complex query support for advanced client integrations.
Sector Modules	Government procurement module; cooperative member management; university grant lifecycle.
Internationalisation	RTL language support; full locale-aware formatting; non-English UI language packs.
Advanced Security	Annual penetration test certification; SOC 2 Type II readiness audit; ISO 27001 alignment.
Marketplace	50+ third-party custom modules available; ecosystem developer portal.
Analytics	Predictive cash flow; risk scenario modelling; workforce planning forecasts.

6.1 Feature-to-Milestone Mapping Summary

Milestone	P0 Features	P1 Features	P2 Features	Personas Served
Milestone 1 — Foundation	All 47 P0 features	Finance, HR, Legal, Risk, Work, API, Deployment P1s	—	P-01, P-02, P-10, P-11, P-12 P-15, P-17, P-18
Milestone 2 — Growth	—	Remaining P1 features (GRC, Sales, Marketing, Analytics, Java client)	SSO, webhooks, banking	P-03 to P-14, P-16
Milestone 3 — Ecosystem	—	—	BI integration, supply chain,	All 18 personas; adds board and sector users

			board, ESG, investment	
Milestone 4 — Intelligence	—	—	P3 GraphQL, international isation	All personas; AI-augmented workflows

7. Product Success Metrics

UME's product success is measured across four dimensions: adoption, outcome delivery, quality, and ecosystem growth. Metrics are collected automatically from deployed instances (with consent) and from user research.

7.1 Adoption Metrics

Metric	Definition	Target (12 months post-M1)	Source
Organisations onboarded	Total paying organisations with ≥1 active user	500+	Platform telemetry
Daily Active Users (DAU)	Users who take at least one action per day	70%+ of licensed users	Session logs
Module adoption breadth	Average number of modules actively used per organisation	≥ 8 modules	Platform telemetry
API usage	Organisations using the REST API or webhook integration	30%+ of organisations	API gateway logs
Custom modules installed	Third-party or custom modules installed from marketplace	50+ unique modules	Marketplace telemetry
Platform consolidation	Orgs reporting UME as their primary operational platform	80%+ at 12 months	Quarterly survey

7.2 Outcome Metrics

Metric	Definition	Target	Measurement
Compliance deadline failures	% of UME-deployed orgs that miss a statutory filing deadline in a 12-month period	< 5%	Annual compliance survey
Time to consolidated financials	Hours from period end to complete consolidated group P&L	< 4 hours (was 3–10 days)	Finance team survey at 6 months
Admin time saved	Hours/week saved on organisational administration vs. pre-UME	≥ 6 hours/week for founders	User time-tracking survey
KRI-to-action lag	Minutes from KRI threshold breach to risk owner notification	< 5 minutes	Platform event log measurement
Employee lifecycle automation	% of onboarding tasks triggered automatically vs. manually	≥ 90% automated	HR process audit at 6 months
Payroll error rate	% of payroll runs requiring manual correction after initial calculation	< 0.5%	Finance audit

7.3 Quality Metrics

Metric	Target	Measurement Frequency

P0/P1 bug escape rate to production	0 P0 bugs; < 2 P1 bugs per release	Per release
Mean Time to Resolution (P0 bugs)	< 4 hours	Per incident
API uptime (production)	≥ 99.9% (standard) / 99.95% (enterprise)	Continuous; monthly report
API P95 latency	< 500ms	Continuous; weekly review
Test coverage — core and kernel	≥ 80%	Per PR; CI enforcement
Security vulnerability SLA	Critical CVE patched within 72 hours; high within 14 days	Per advisory
Documentation completeness	All P0 features have user-facing documentation before release	Pre-release checklist

7.4 Ecosystem Metrics

Metric	Target (M3 — Q1 2027)	Target (M4 — Q2 2027)
Marketplace modules published	20+	50+
Registered SDK developers	100+	500+
Jurisdiction policy packs active	45+	65+
Community forum members	500+	2,000+
Partner integrations (native connectors)	Slack, Teams, Salesforce, QuickBooks, Xero	+ SAP, NetSuite, HubSpot, Stripe

8. Constraints, Assumptions & Dependencies

8.1 Product Constraints

Constraint	Impact on Product Decisions
Must be operable by a non-technical 1-person business without IT staff	Embedded mode must be truly zero-configuration; all setup through UI; no TOML editing required for standard use
Must comply with GDPR, CCPA, and equivalent data residency requirements	Data export, deletion, and residency controls are P0; field-level PII encryption is mandatory
Core operations must function offline (embedded mode)	All P0 features must work without network connectivity; sync model for future cloud mode
Must be affordable for a 1-person organisation	Pricing model must include a solo-accessible tier; platform cost must be < competitive tool stack cost at every scale
Jurisdiction policy packs must be kept current with regulatory changes	Requires a separate content/regulatory team operating post-launch; pack updates ship independently from platform releases
Must not require a Rust developer to operate or extend the platform	All configuration via UI or env vars; SDK designed for module developers, not platform operators

8.2 Assumptions

ID	Assumption	Risk if Wrong
A-01	Organisations will trust a single platform with all their operational data	Security and on-premises deployment must be demonstrably strong; self-hosted option is a M1 requirement
A-02	PostgreSQL and SQLite are sufficient database backends for Milestone 1–2 scale	May need to prioritise additional backends (CockroachDB, Aurora) earlier if large enterprises adopt faster than expected
A-03	The Rust ecosystem provides sufficient library maturity for all required integrations (SMTP, S3, JWT, Argon2)	Tracked per dependency; alternatives selected in advance for critical paths
A-04	Jurisdiction compliance obligations can be modeled as versioned rule-based policy packs for 45+ jurisdictions	High-complexity jurisdictions (US federal + 50-state, EU multi-framework) may require expert legal review per pack
A-05	Third-party developers will adopt the module SDK if it is well-documented and discoverable	Requires investment in developer documentation, sample modules, and active community engagement from launch
A-06	Banking open banking APIs are stable enough for automated bank feed reconciliation in key markets (UK, EU, AU)	Bank API reliability varies; manual import fallback must always be available

8.3 External Dependencies

Dependency	Type	Milestone	Risk Level	Mitigation
PostgreSQL 16+	Required for Standard/Enterprise deployment	M1	Low	Stable, well-supported; SQLite fallback always available

NATS JetStream or Kafka	Required for multi-node event bus	M1	Medium	In-memory bus available for single-node; external broker optional
Open Banking APIs	Required for bank feed reconciliation	M2	High	Manual bank statement import is the primary path; API integration is enhancement
Identity providers (Azure AD, Okta)	Required for enterprise SSO	M2	Low	Local auth is P0 and always available; SSO is enhancement
Tax data provider	Required for real-time payroll tax rates in 45+ jurisdictions	M1	Medium	Built-in tax tables updated per release; live API integration deferred to M2
S3-compatible storage	Required for document management and backup off-site replication	M1 (backup)	Low	Multiple providers (AWS S3, GCS, MinIO); self-hosted option available
Java 17+ runtime	Required for Java client suite	M2	Low	LTS version; widely available

8.4 Out of Scope (v1.0)

The following are explicitly out of scope for UME v1.0 and will be considered for future versions:

- Mobile native applications (iOS/Android) — the web portal is responsive; native apps are post-v1.0
- AI/LLM-generated reports or autonomous actions — AI deal scoring and anomaly detection (M4) are the extent of AI in v1.0
- Inter-organisation data sharing or network effects — UME v1.0 is single-organisation; B2B collaboration features are post-v1.0
- Real-time accounting validation with tax authorities (Making Tax Digital streaming API) — standard submission flows only in v1.0
- Built-in video conferencing or document co-editing — integrations with Zoom, Google Docs etc. are in scope; building these is not
- Blockchain-based audit trail — the cryptographic hash chain provides tamper evidence; distributed ledger is not required for this use case

Appendix A: Module Catalogue

Complete listing of all 42 UME organisation modules with their identifiers and primary capabilities.

Mod ID	Module Name	Domain	Primary Capability	Milestone
MOD-01	Administration	Organisation	Org structure, policies, calendar, delegation	M1
MOD-02	Analytics	Intelligence	Cross-domain KPIs, dashboards, trend analysis	M2
MOD-03	Backup	Infrastructure	Scheduled backup, restore, encryption, off-site replication	M1
MOD-04	Board	Governance	Board meetings, resolutions, packs, quorum, conflict	M3
MOD-05	Business Development	Growth	Partnerships, BD pipeline, MOU tracking	M3
MOD-06	Content Management	Knowledge	Documents, versioning, approval workflow	M2
MOD-07	Communications	Platform	Email, SMS, push, in-app notification dispatch	M1
MOD-08	CRM	Revenue	Accounts, contacts, interactions, relationships	M2
MOD-09	Design System	Platform	UI theme, component library, branding configuration	M1
MOD-10	Engineering	Technology	Codebase, CI/CD pipeline visibility, tech radar	M3
MOD-11	ESG	Governance	Emissions tracking, ESG reporting frameworks	M3
MOD-12	Enterprise Management	Administration	Enterprise-wide configuration, multi-instance admin	M2
MOD-13	Legal Entity (Chombo)	Compliance	Entity registry, jurisdiction packs, filing calendar	M1
MOD-14	Finance	Finance	Ledger, AR/AP, payroll, multi-entity, statements	M1
MOD-15	GRC	Compliance	Obligations, controls, audits, evidence collection	M2
MOD-16	Human Resources	People	Lifecycle, compensation, leave, payroll, performance	M1
MOD-17	Investment	Finance	Fund management, LP accounts, portfolio, IC governance	M3
MOD-18	IT & Assets	Technology	CMDB, ITSM, change management, software licensing	M2
MOD-19	Knowledge	Collaboration	Knowledge base, search, article management	M2
MOD-20	Learning & Development	People	LMS, learning paths, compliance training, certificates	M2
MOD-21	Strategy	Leadership	OKRs, strategic initiatives, strategic review cycles	M2
MOD-22	Marketing (Soko)	Revenue	Campaigns, strategy packs, signal ingestion, attribution	M2
MOD-23	Master Data Management	Platform	Golden record management, deduplication, MDM workflow	M3

MOD-24	Office & Facilities	Operations	Space management, visitor management, desk booking	M3
MOD-25	Operations	Operations	Operational KPIs, cross-departmental processes	M2
MOD-26	Portal	Platform	Employee portal, search, notification centre, quick actions	M1
MOD-27	Portfolio	Projects	Portfolio, programme, project hierarchy; milestones	M2
MOD-28	PR & Branding	Communications	Press management, brand guidelines, media contacts	M3
MOD-29	Process & Orchestration	Platform	Workflow engine, process templates, SLA management	M1
MOD-30	Product Management	Technology	Product backlog, roadmap, feedback, release planning	M3
MOD-31	Production & Manufacturing	Operations	Production orders, BOMs, quality control, MES integration	M3
MOD-32	Requirements Management	Technology	Business and technical requirements, traceability matrix	M3
MOD-33	Risk Management	Governance	Risk register, KRIs, treatment plans, heat map	M1
MOD-34	Sales	Revenue	Pipeline, forecasting, commissions, territory, AI scoring	M2
MOD-35	Schedule	Operations	Organisational calendar, resource scheduling, booking	M2
MOD-36	Security	Compliance	IAM, incidents, vulnerabilities, privacy, vault, SIEM	M1
MOD-37	Supply Chain	Operations	Procurement, inventory, supplier management	M2
MOD-38	Teams & Cooperatives	People	Team management, cooperative member equity, governance	M3
MOD-39	Templates	Platform	Document and workflow template library	M2
MOD-40	Work Management	Delivery	Work items, sprints, backlogs, time logging	M1
MOD-41	Custom UME Modules	Extension	Platform-provided custom module framework	M1
MOD-42	Custom Org Modules	Extension	Organisation-private custom module framework	M1

Appendix B: User Story Index

Story ID	As a…	I want to…	Module	Milestone
US-ADM-001	P-01 CEO	manage the org entity structure	MOD-01	M1
US-ADM-002	P-06 Legal Officer	publish a policy and track acknowledgment	MOD-01	M1
US-FIN-001	P-02 CFO	see consolidated group financials in real time	MOD-14	M2

US-FIN-002	P-11 Finance Manager	process a supplier invoice via 3-way match	MOD-14	M1
US-FIN-003	P-11 Finance Manager	run payroll with automatic journal posting	MOD-14	M1
US-FIN-004	P-17 Freelancer	send an invoice and track payment	MOD-14	M1
US-CHM-001	P-06 Legal Officer	see all upcoming filing deadlines across all entities	MOD-13	M1
US-CHM-002	P-01 Founder	register a new entity and get a compliance picture	MOD-13	M1
US-HR-001	P-12 HR Manager	onboard a new employee with automated tasks	MOD-16	M1
US-HR-002	P-15 Employee	request leave and see my balance	MOD-16	M1
US-HR-003	P-05 CHRO	run an annual performance review cycle	MOD-16	M2
US-GRC-001	P-07 Risk Manager	be alerted automatically when a KRI breaches a threshold	MOD-33	M2
US-GRC-002	P-13 Compliance Manager	prepare for an internal audit without 2 weeks of prep	MOD-15	M2
US-SAL-001	P-10 VP Sales	forecast revenue accurately for the next quarter	MOD-34	M2
US-SAL-002	P-10 VP Sales	calculate and distribute commission statements	MOD-34	M2
US-SEC-001	P-08 CTO	respond to a P1 security incident with a structured process	MOD-36	M2
US-IT-001	P-16 IT Admin	manage a production change through a full CAB process	MOD-18	M2
US-WRK-001	P-14 Project Manager	run a sprint with full visibility into delivery progress	MOD-40	M1
US-WRK-002	P-15 Employee	find any info and take action from a single portal	MOD-26	M1

Appendix C: Acceptance Criteria Template

All user stories must have acceptance criteria documented before sprint planning. Use the following template:

AC Template

```
Given [context / precondition]
When  [action taken by the user or system]
Then  [expected observable outcome]
And   [additional expected outcomes, if any]

Non-functional criteria:
  Performance: [response time / throughput target, if applicable]
  Audit:       [what audit record must be created]
  RBAC:        [what permission is required]
  Event:       [what domain event must be published]
```

Appendix D: Glossary

Term	Definition
Chombo	The UME Legal Entity Management module. Chombo is the Swahili/Shona word for "tool" or "instrument".
Soko	The UME Marketing module. Soko is the Swahili word for "market".
Kernel	The UME runtime process composition root: manages all modules, subsystems, and shared infrastructure.
Policy Pack	A versioned, machine-readable set of compliance obligations for a specific jurisdiction and entity type.
Domain Event	An immutable fact emitted to the event bus after a state-changing operation, in format {domain}.{resource}.{verb}.
KRI	Key Risk Indicator: a metric that provides early warning of a risk approaching its appetite threshold.
SUS	System Usability Scale: a 10-question standardised questionnaire producing a 0–100 usability score.
JTBD	Jobs To Be Done: the underlying goal or progress a user is trying to make by using the product.
P0/P1/P2/P3	Product priority levels: P0 = must-have for launch, P1 = required for GA, P2 = important, P3 = desirable.
Multi-tenancy	The ability to serve multiple organisations from a single platform deployment with complete data isolation.
RFC 2119	IETF document defining the requirement level keywords SHALL, SHOULD, MAY used in UME specifications.
RBAC	Role-Based Access Control: permissions are assigned to roles, roles are assigned to users.
CAB	Change Advisory Board: governance body that reviews and approves IT changes before implementation.
DSAR	Data Subject Access Request: a request under GDPR/CCPA for an individual to see their personal data.
RoPA	Record of Processing Activities: a mandatory GDPR register of all personal data processing activities.
DPIA	Data Protection Impact Assessment: a structured risk assessment required for high-risk data processing.